hyper2kvm

Enterprise VM Migration Platform

Production-grade toolkit for migrating VMware, Hyper-V, and Azure workloads to KVM/libvirt and Kubernetes — with zero downtime and full automation.

Proprietary (HyperSDK) — Built for Enterprise Linux

The Problem

Why enterprises are leaving VMware — and what's at stake.

💸

Broadcom Licensing Shock

VMware license costs increased 2-12x after the Broadcom acquisition. Perpetual licenses eliminated. Bundled pricing forces customers to pay for unused products.

🔒

Vendor Lock-in Risk

VMDK, vSphere APIs, and proprietary tooling create deep dependencies. Every year of delay increases switching costs exponentially.

⚠️

End-of-Support Pressure

Legacy vSphere versions losing support. Security patches stop. Compliance audits flag unpatched hypervisors. Migration becomes urgent, not optional.

🏗️

Cloud-Native Gap

VMware workloads can't natively run in Kubernetes. Modernization requires re-platforming VMs to KubeVirt — but manual migration is slow and error-prone.

> *"Organizations face a stark choice: absorb 300-1200% cost increases, or migrate to open alternatives. The window for planned migration is closing."*

Cost Impact

The financial case for migration is overwhelming.

$0
KVM/libvirt License Cost (vs $5,000+/socket/yr VMware)

70%
Average TCO Reduction over 3 years

10x
ROI on Migration Investment

3-Year TCO Comparison (100 VMs)

Cost Category	VMware vSphere	KVM + KubeVirt	Savings
Hypervisor Licensing	$500,000	$0	**$500,000**
Management Platform	$200,000	$0 (Cockpit/CLI)	**$200,000**
Support Subscription	$150,000	$50,000 (RHEL optional)	**$100,000**
Migration Tooling	N/A	Contact for pricing* (hyper2kvm)	**$0**
Staff Training	$30,000	$20,000	**$10,000**
Total	**$880,000**	**$70,000**	**$810,000**

The Solution

hyper2kvm — one tool for every migration path.

VMware vSphere → hyper2kvm → KVM / libvirt

Hyper-V / VHD → hyper2kvm → KubeVirt / K8s

Azure / AWS → hyper2kvm → On-Prem KVM

Any Source Format

VMDK, OVA, OVF, VHD/VHDX, RAW, AMI — auto-detected. No manual format juggling.

Offline Guest Fixes

Automatic fstab repair, initramfs rebuild, driver injection, GRUB fix — VMs boot on first try.

Dual Deploy Target

Deploy to libvirt (traditional) or KubeVirt (cloud-native) from the same pipeline. Or both.

Terminal UI (zkvm)

Interactive TUI with vSphere discovery, VM selection, live progress, file browser — no CLI memorization needed.

Platform Capabilities

Enterprise features built for production workloads.

Capability	hyper2kvm	virt-v2v	Manual
VMware vSphere integration (govc/pyvmomi)	✓	✓	✗
Hyper-V / VHD support	✓	✓	✗
Azure VM export	✓	✗	✗
KubeVirt / Kubernetes deploy	✓	✗	✗
Interactive TUI (guided wizard)	✓	✗	✗
Windows driver injection	✓	✓	✗
Automatic fstab/GRUB repair	✓	Partial	✗
Live progress streaming	✓	✗	✗
Batch / queue migration	✓	✗	✗
YAML-driven automation	✓	✗	✗
Pre-flight validation	✓	✗	✗
Boot verification test	✓	✗	✗
Proprietary (HyperSDK)	✓	✓ **(GPL)**	N/A

Migration Workflow

From VMware to production KVM in 4 steps.

1. **Discover & Select**
   Connect to vCenter, discover all VMs with CPU/RAM/power state. Select targets interactively or via YAML. Pre-flight checks validate the environment.

2. **Export & Convert**
   Export VMs via govc/OVF/HTTPS with live progress. Convert VMDK→qcow2 with compression and flattening. Supports multi-disk VMs and snapshots.

3. **Fix & Optimize**
   Offline guest repairs: fstab stabilization (UUID/path), initramfs rebuild with virtio drivers, VMware tools removal, SELinux relabel, GRUB reconfiguration.

4. **Deploy & Verify**
   Generate libvirt domain XML, register with virsh, or deploy to KubeVirt on Kubernetes. Automatic boot test validates the migrated VM starts correctly.

⚡

Minutes, Not Days

Typical VM migration: 5-30 minutes depending on disk size. No manual intervention required for standard Linux/Windows guests.

🔁

Batch Processing

Queue multiple VMs for sequential migration. Track progress per-VM. Failed VMs don't block the queue — results summarized at completion.

Why KubeVirt?

Run VMs and containers on the same platform — the best of both worlds.

☸️

Unified Platform

VMs and containers managed by Kubernetes. One API, one toolchain, one team.

📈

Cloud-Native Ops

GitOps, CI/CD, Helm charts, operators — apply modern practices to VM workloads.

🔄

Gradual Modernization

Migrate VMs first, containerize later. No big-bang rewrite. Incremental path to cloud-native.

💰

No Hypervisor Tax

KubeVirt runs on standard Linux + KVM. No per-socket licensing. Scale with commodity hardware.

🌍

Multi-Cloud Ready

Same KubeVirt manifests work on-prem, in AWS (EKS), Azure (AKS), or GCP (GKE). True portability.

🛡️

Enterprise Backed

KubeVirt is a CNCF project, backed by Red Hat (OpenShift Virtualization). Production-proven at scale.

> *"KubeVirt lets you run your legacy VMs in Kubernetes today, and modernize them into containers tomorrow — on your schedule, not your vendor's."*

Supported Platforms

Comprehensive source and target compatibility.

Migration Sources

- VMware vSphere 6.x / 7.x / 8.x
- VMware ESXi (standalone)
- Microsoft Hyper-V
- Microsoft Azure VMs
- AWS EC2 (AMI export)
- Raw disk images
- Any VMDK/OVA/OVF/VHD file

Deploy Targets

- KVM / libvirt (RHEL, Fedora, Ubuntu)
- KubeVirt on Kubernetes
- K3s + KubeVirt (edge)
- OpenShift Virtualization
- Proxmox VE (qcow2 import)
- Any KVM-based hypervisor

Guest OS Support

- RHEL 7 / 8 / 9 / 10
- CentOS, AlmaLinux, Rocky
- Ubuntu 18.04–24.04
- Debian 10–13
- SUSE / openSUSE
- Windows Server 2016–2025
- Windows 10 / 11
- Photon OS, Fedora, Arch

Output Formats

- qcow2 (default, compressed)
- RAW (maximum performance)
- VDI (VirtualBox compatible)
- Libvirt domain XML
- KubeVirt VirtualMachine YAML

Customer Scenarios

Real-world migration patterns hyper2kvm solves.

🏢

VMware Exit — Enterprise

Scenario: 500 VMs on vSphere, facing 5x license increase.
Solution: Batch migrate to RHEL + KVM using hyper2kvm YAML automation. Queue processing handles overnight migrations. Boot verification catches issues automatically.
Result: $2M+ annual savings, zero vendor lock-in.

☁️

Cloud Repatriation

Scenario: Azure VMs costing $50K/month, moving to on-prem.
Solution: Export Azure VMs via hyper2kvm's Azure integration. Convert VHD→qcow2, deploy to local KVM cluster.
Result: 60% cloud cost reduction, data sovereignty maintained.

🚀

Kubernetes Modernization

Scenario: Legacy VMs that can't be containerized yet, but team wants Kubernetes.
Solution: Migrate VMs to KubeVirt. Run alongside containers. Modernize incrementally.
Result: Unified platform, single operations team, gradual containerization path.

🏭

Edge / Factory Floor

Scenario: Industrial VMs running on expensive VMware licenses at remote sites.
Solution: K3s + KubeVirt on lightweight hardware. hyper2kvm converts and deploys.
Result: Zero license cost at edge, GitOps management from central cluster.

Why hyper2kvm?

What sets us apart from alternatives.

🎯

First-Boot Success

Offline guest fixes ensure VMs boot on first try. No manual driver installation, no broken fstab, no missing initramfs. We fix it before you notice.

🔧

Production-Grade

1,273 unit tests. Pre-flight validation. Structured logging with tracing. Error diagnosis. This isn't a script — it's a platform.

📊

Interactive TUI

The zkvm terminal UI provides guided migration with live progress, file browser, vSphere discovery, and deploy toggles. No CLI memorization needed.

📝

YAML Automation

Every migration is reproducible via YAML config. Perfect for CI/CD pipelines, GitOps workflows, and batch processing. Automate hundreds of VMs.

Ready to Eliminate VMware Costs?

hyper2kvm is Proprietary (HyperSDK) and ready for production.
Start migrating today — no license required, no vendor approval needed.

github.com/ssahani/hyper2kvm

Getting Started

From zero to migrated VM in 2 commands.

Quick Install

```
# Install
sudo ./scripts/quickstart.sh

# Migrate a VMDK
sudo h2kvmctl --cmd local --vmdk disk.vmdk --output-dir ./out

# Or use the interactive TUI
./zkvm/zkvm

# Or use YAML automation
sudo h2kvmctl --config migration.yaml
```

Example YAML Config

```yaml
cmd: vsphere
vcenter: 10.0.0.100
vc_user: administrator@vsphere.local
vc_password_env: VC_PASSWORD
dc_name: Datacenter1
vs_action: export_vm
vs_vm: production-web-01
output_dir: ./out
out_format: qcow2
emit_domain_xml: true
regen_initramfs: true
deploy_k8s: true
k8s_vm_name: web-01
k8s_namespace: production
```

1,273 Unit Tests Passing	**Proprietary (HyperSDK)**	**Python + Go**

Open Source License

Dual Language Platform